

1st International Workshop on Real and Virtual Spaces Influences (ReViSI)

Laura Battistel*
University of Trento
Eurac Research

Kangsoo Kim†
University of Calgary

Riccardo Parin‡
Eurac Research

Massimiliano Zampini§
University of Trento

Gerd Bruder¶
University of Central Florida

Gregory F. Welch||
University of Central Florida



Figure 1: Graphical representation of the central relationship explored in ReViSI—exploring the interplay of real and virtual worlds.

ABSTRACT

The International Workshop on Real and Virtual Spaces Influences (ReViSI) promotes a multidisciplinary dialogue on the interplay between real and virtual environments, addressing the central question: how do these domains influence one another? The workshop seeks to advance understanding of whether real-world stimuli can enhance presence in Extended Reality (XR) and how virtual stimuli may reshape perceptions of the physical world. Topics of interest include enhancing presence and social presence in XR through congruent real-virtual environments, identifying best practices for integrating real-world elements to enrich XR experiences, and exploring the potential of real-world stimuli to improve, for instance, training and rehabilitation outcomes in virtual reality. Additionally, the workshop examines how virtual stimuli can alter real-world perception and identifies sensory modalities most affected by virtual stimuli. By addressing these themes, the workshop aims to contribute novel insights to the growing field of real-virtual synergy and its applications.

Index Terms: Multisensory interaction, presence, environment.

1 INTRODUCTION

The inaugural 1st International Workshop on Real and Virtual Spaces Influences (ReViSI) aims to foster multidisciplinary dialogue addressing a pivotal question: *how do real and virtual environments influence each other?* This workshop brings together researchers from fields such as Extended Reality (XR), cognitive

psychology, and beyond, with a shared interest in exploring the dynamic interplay between physical and virtual spaces. A significant focus of ReViSI is understanding whether real-world stimuli can enhance presence and social presence in XR, and how virtual stimuli might reshape perception of the real world (Fig. 1). These questions are critical in XR research, where creating ecologically valid and immersive experiences has practical implications for various application areas, e.g., training, rehabilitation, and entertainment.

Research in the XR community has long focused on enhancing the concept of presence, defined as the “perceptual illusion of non-mediation” [10]. Additionally, virtual stimuli have been shown to influence real-world perceptions, such as altering taste or internal bodily states, and XR training has demonstrated potential for real-world generalization [2, 3, 4].

Despite these advancements, findings on multisensory integration in XR remain inconsistent [1, 9, 12]. Some studies highlight significant benefits, while others report minimal or inconsistent effects, often due to variations in study design, measurement methods, or stimulus delivery [6, 5, 11]. These challenges raise critical questions central to ReViSI’s goals:

- How can congruent real-virtual environments reliably enhance (social) presence in XR, and what are the underlying mechanisms?
- What are the best practices for integrating real-world stimuli into XR experiences?
- How might virtual stimuli alter users’ real-world perceptions, or vice versa?

Addressing these issues requires establishing standardized methodologies and exploring the bidirectional relationship between real and virtual environments, which are the workshop’s objectives.

2 CHALLENGES

One of the primary challenges faced by the XR community is establishing a unified and decisive definition of presence, along with effective methods for its measurement [7]. In fact, traditional measurement approaches, such as questionnaires, are highly subjective and prone to recall biases [13]. While physiological or neurological

*e-mail: laura.battistel@eurac.edu

†e-mail: kangsoo.kim@ucalgary.ca

‡e-mail: riccardo.parin@eurac.edu

§e-mail: massimiliano.zampini@unitn.it

¶e-mail: bruder@ucf.edu

||e-mail: welch@ucf.edu

data could provide more objective insights, their application is often constrained by intrinsic limitations, such as the restricted range of movement they permit [8].

Our workshop, while promoting dialogue on the interplay between virtual and real environments, also seeks to encourage reflection on this critical issue. By addressing these challenges, we aim to inspire new ideas on how to define, measure, and enhance presence in XR.

3 CALL FOR PAPERS

ReViSI invited submissions of original papers (up to 6 pages, excluding references) in English, prepared in IEEE Computer Society VGTC format. We welcomed *position papers* sharing insightful ideas, *survey papers* highlighting research gaps and future directions, *project papers* outlining ongoing or planned projects, and *research papers* presenting state-of-the-art advances with supporting results. Accepted papers are published in the IEEE Digital Library as part of the Proceedings of the IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (IEEE VRW). Submissions underwent a single-blind review by workshop organizers, with constructive feedback provided.

4 WORKSHOP ORGANIZATION

ReViSI will take place as a half-day in-person workshop at IEEE VR 2025, running from 14:00 to 18:00 on March 8, 2025. The schedule includes an introduction, engaging invited talks offering in-depth insights into the workshop theme, and a paper presentation session showcasing the latest research contributions and presenting discussion points. The workshop aims to foster active interaction and idea exchange among participants, culminating in a summary and closing session to reflect on key outcomes. A tentative schedule is shown in Tab. 1.

5 KEYNOTE TALKS

Three experts are invited to deliver talks at our workshop and subsequently contribute to the discussion:

- **Tabitha C. Peck** (Associate Professor of Mathematics and Computer Science at Davidson College) will discuss how users' previous experiences and backgrounds influence their perception of virtual environments (VEs) and, in turn, how VEs may affect user behavior.
- **Tanja Kojic** (Senior User Experience (UX) Researcher) will explore how stimuli from the real environment, such as sounds and noises, impact users' experiences within a VE.
- **Christopher Dawes** (Research Fellow at University College London) will examine how the act of eating within a VE affects users' taste perception.

6 PAPER PRESENTATIONS

Four papers have been accepted for presentation at our workshop:

- Nöring et al. present a study demonstrating how congruent haptic feedback during fire training in a VE can be both effective and reliable.
- Waligórski et al. focus on the often-overlooked role of users' real-world environments and backgrounds in the field of digital dualism, highlighting their importance for creating more effective XR experiences.
- Furuya et al. provide a theoretical discussion on the relationship between presence and trust in XR, proposing ways to enhance presence to improve interactions with virtual agents.
- Khan et al. describe how spatial navigation training in a VE closely mimicking the real world can generalize its effects to physical spaces.

Table 1: Tentative workshop schedule.

Time	Content
14:00–14:20	Introduction and Greetings
14:20–15:35	Keynote Talks
15:35–15:50	Break
15:50–16:40	Panel Discussion
16:40–16:55	Break
16:55–17:55	Paper Presentations
17:55–18:00	Closing

7 CONCLUSION

The inaugural edition of the ReViSI aims to promote meaningful discussions on how real and virtual environments influence each other. The workshop seeks to gather insights on how to employ congruent real-virtual stimuli to enhance presence and improve training and rehabilitation outcomes. Additionally, it investigates how virtual experiences can shape users' perceptions of the real world.

REFERENCES

- [1] I. d. V. Bosman, O. O. Buruk, K. Jørgensen, and J. Hamari. The effect of audio on the experience in virtual reality: a scoping review. *Behaviour & Information Technology*, 43(1):165–199, 2024. 1
- [2] P. Burkert et al. Colortable: Manipulating tasting experiences, impact of light color on food flavor perception. In *International Conference on Human-Computer Interaction*, pages 595–601, 2022. 1
- [3] A. A. Chang, E. Kazemi, V. Esmaili, and M. S. Davies. The effectiveness of virtual reality training: A systematic review. *Journal of Organizational Behavior Management*, 44(3):214–232, 2024. 1
- [4] A. El Ali, R. Ney, Z. M. Van Berlo, and P. Cesar. Is that my heart-beat? measuring and understanding modality-dependent cardiac interoception in virtual reality. *IEEE Transactions on Visualization and Computer Graphics*, 29(11):4805–4815, 2023. 1
- [5] M. Feng, A. Dey, and R. W. Lindeman. An initial exploration of a multi-sensory design space: Tactile support for walking in immersive virtual environments. In *Proceedings of the IEEE Symposium on 3D User Interfaces*, pages 95–104, 2016. 1
- [6] S. Günther, A. Skogseide, R. Buhlmann, and M. Mühlhäuser. Assessing the influence of visual cues in virtual reality on the spatial perception of physical thermal stimuli. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*, pages 1–12, 2024. 1
- [7] D. Hein, C. Mai, and H. Hußmann. The usage of presence measurements in research: a review. In *International Society for Presence Research Annual Conference*, pages 21–22, 2018. 1
- [8] F. Honegger, Y. Feng, and G. M. Rauterberg. Multimodality for passive experience: effects of visual, auditory, vibration and draught stimuli on sense of presence. *Journal of Universal Computer Science*, 27(6):582–608, 2021. 2
- [9] H. Khan and D. Nilsson. Smell of fire increases behavioural realism in virtual reality: A case study on a recreated mgm grand hotel fire. In *Proceedings of the IEEE International Symposium on Mixed and Augmented Reality*, pages 820–828, 2023. 1
- [10] M. Lombard. At the heart of it all: The concept of presence. *Journal of Computer-Mediated Communication*, 3(2), 1997. 1
- [11] M. Melo, H. Coelho, G. Gonçalves, N. Losada, F. Jorge, M. S. Teixeira, and M. Bessa. Immersive multisensory virtual reality technologies for virtual tourism: A study of the user's sense of presence, satisfaction, emotions, and attitudes. *Multimedia Systems*, 28(3):1027–1037, 2022. 1
- [12] E. Sikström et al. Shop'til you hear it drop: influence of interactive auditory feedback in a virtual reality supermarket. In *Proceedings of the 22nd ACM Conference on Virtual Reality Software and Technology*, pages 355–356, 2016. 1
- [13] J. Van Baren. Measuring presence: A guide to current measurement approaches. *Deliverable of the OmniPres project IST-2001-39237*, 2004. 1